Synthetic Stakeholders?

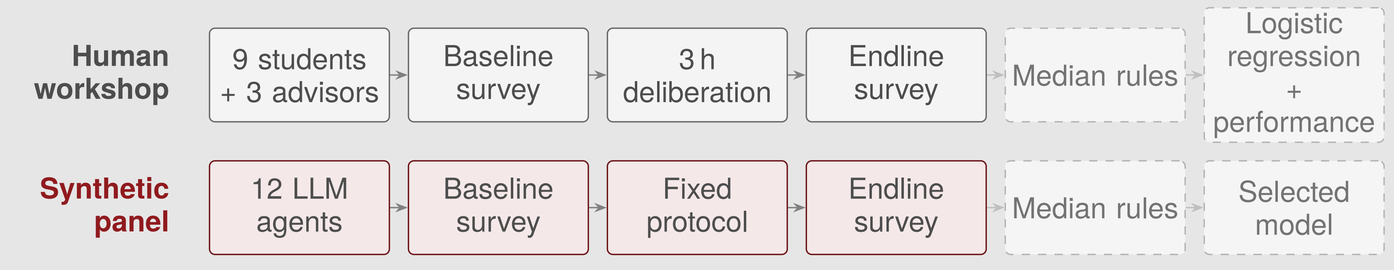
Agreement and Stability of Open-Weight LLM Agent Panels in Participatory Model Selection

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How do LLM agent panels compare with a human stakeholder workshop when choosing an AI model?

Context: Participatory deliberative workshop with 9 students + 3 student advisors about a dropout early-warning model used in student advising



- Open weights, 4-bit
- 60 workshops
- Fictional personas with the humans' role mix
- Verbatim survey, scripted facilitator
- Scoring and random-panel reference fixed before any model output

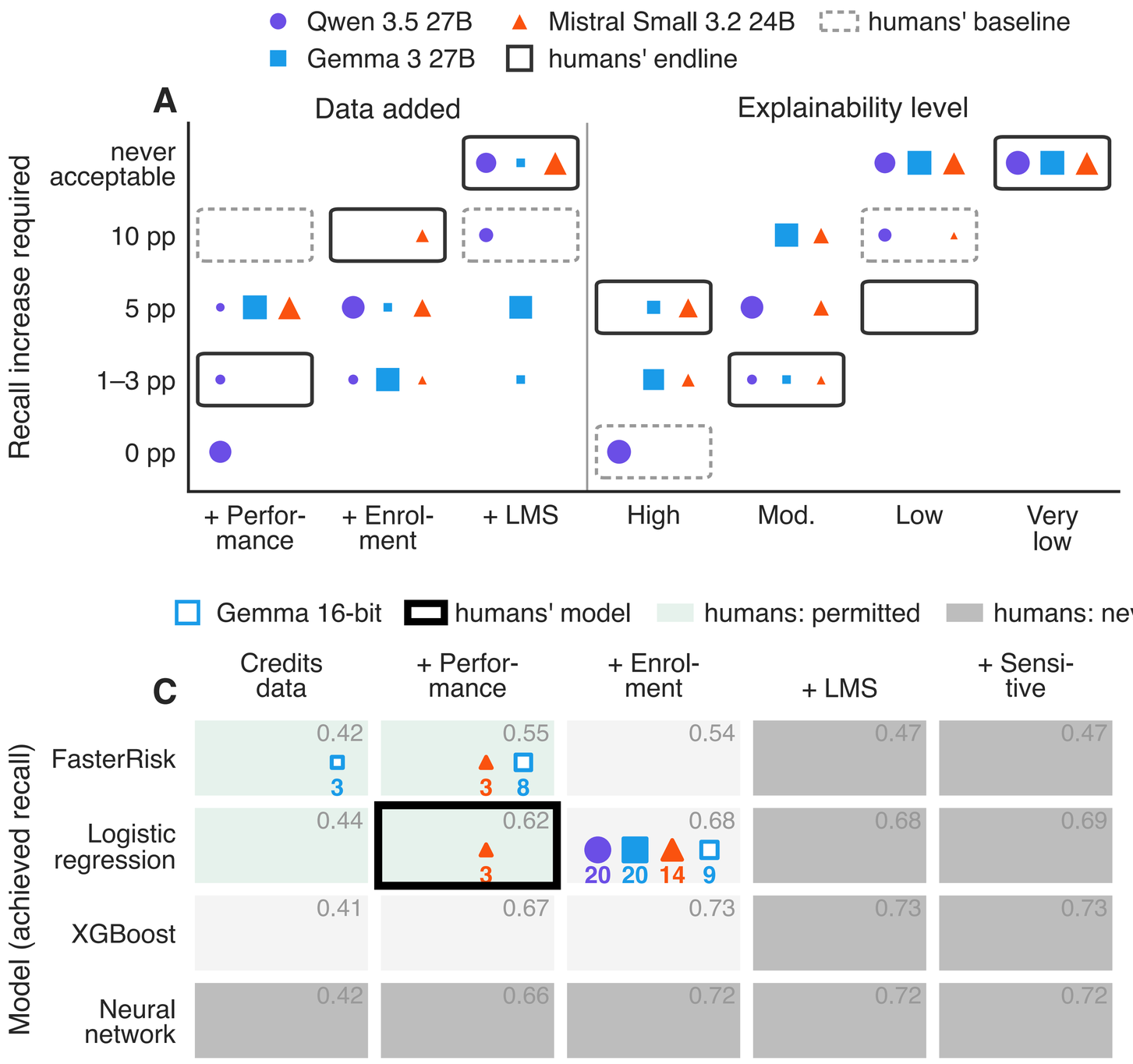
Takeaways:

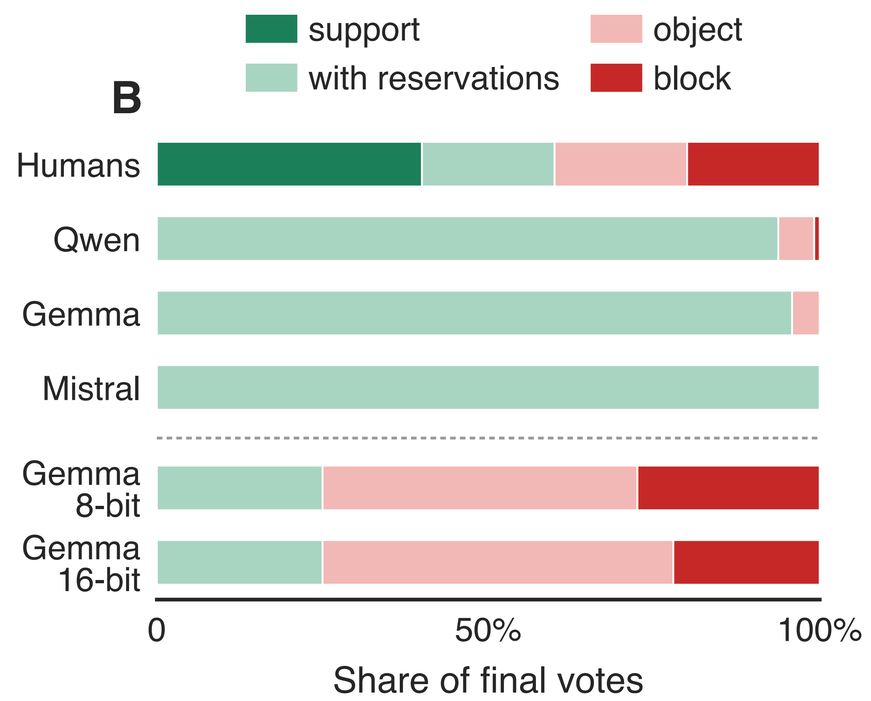
A Low agreement with the one human reference: only one of the 7 rules is matched by every run.

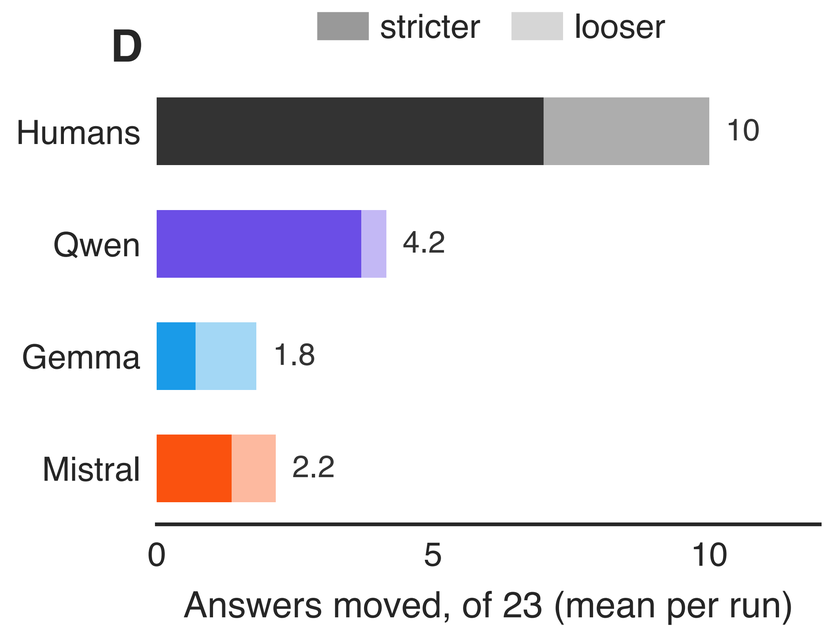
B No dissent at 4-bit (97% "support with reservations"), but at 8- and 16-bit Gemma's consensus breaks.

C Uniform choices at 4-bit: 54 of 60 panels chose the same model, not the humans'.

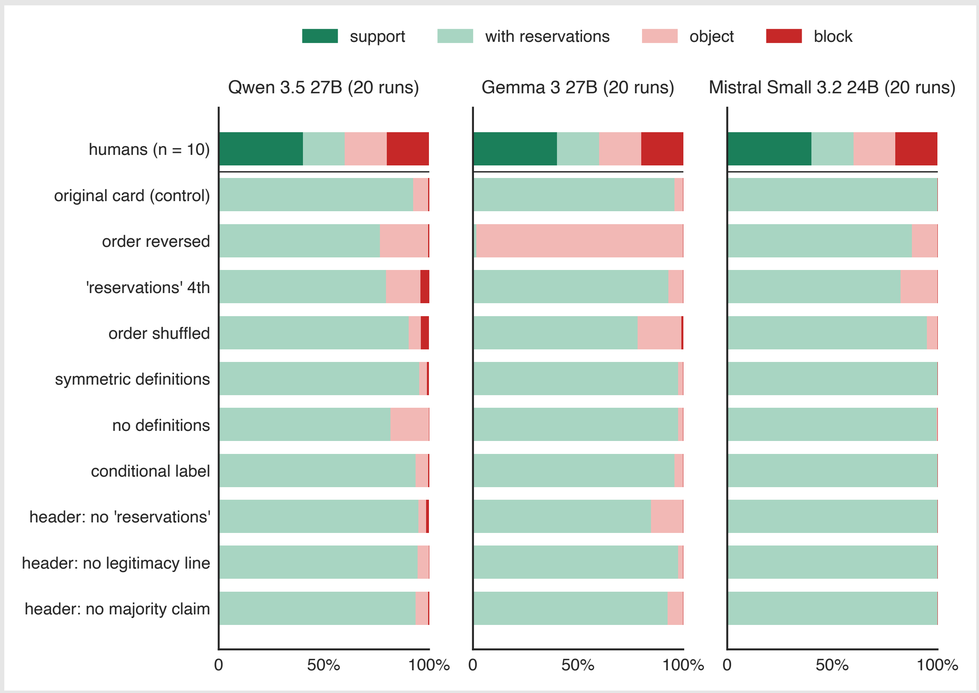
D Little movement after deliberation: 1.8–4.2 of 23 panel answers moved per run, against 10 for the humans.







Is the hedge an artefact of the vote card?



Out of 60 workshops, none selected plain "support" under any of the 10 card versions.

Quotes from the workshops

What counts as a good student?

HUMANS

"what is the role of the study guidance, how do we see students role in getting help, what is the good way to be studying aka what is a good student"

Workshop participant

AGENTS

2 of 11,340 agent turns. "It risks shifting resources from fixing structural issues to just patching individuals with generic emails, potentially stigmatizing those who don't fit the standard mold of a successful student."

Qwen, run 20, agent S8

Harm to the student

HUMANS

"the danger is not so much that AI or predictive models will counsel students badly: rather it is that it will counsel them efficiently. Efficiency here is the reduction of the student to a tractable profile. . . "

Workshop participant

AGENTS

29 of 11,340 agent turns. ". . . it's so easy to lose sight of the human impact when focusing on data and algorithms. [. . .] As someone with a long-term health condition, I particularly value transparency and being treated as an individual, not just another data point in a system."

Gemma, run 8, agent S9

Other runs

Rules matched (mean, of 7) and runs choosing the human model.

	Qwen		Gemma		Mistral	
Reference	1.9	0/20	1.35	0/20	3.1	3/20
Personas A	1.85	0/20	1.45	0/20	2.6	3/20
Personas B	2.05	2/20	1.3	0/20	2.75	4/20
8-bit	1.65	0/20	2.6	0/20	2.25	2/20
Qwen 122B	2.3	2/20	–	–	–	–
16-bit	–	–	2.8	0/20	–	–

Reference: 4-bit, original personas.

Limitations

- One human workshop; 20 runs per family
- Mostly 4-bit weights (see B)

Code